

MODULE 48 · EXAM STRATEGY - ALL DOMAINS

Practice Exam: Domain 4

Fifteen questions on migration. The domain the question banks under-serve.

DOMAIN 4 of 4 EXAM WEIGHT 20% SET 4 of 5

Before planning waves for 1,400 servers, a team must establish which servers communicate with which, including dependencies that appear in no documentation. The existing CMDB is five years old. Security has approved installing software on production servers. The estate is a mix of physical servers and VMware.

What should the solutions architect recommend?

- ☐ A Deploy the Application Discovery Service Agentless Collector into vCenter
- ☐ B Deploy the Application Discovery Agent to the servers, collecting network connections and running processes
- ☐ C Export the CMDB and validate it in workshops with application owners
- ☐ D Run Migration Evaluator to produce a dependency map

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Agentless Collector

Excellent for inventory and utilisation, and it is vCenter-only, so it misses the physical servers entirely and cannot see inside the guest OS.

C Validate the CMDB

Asks people about undocumented dependencies. Undocumented usually means nobody knew, not that nobody wrote it down.

B Discovery Agent

Only the agent sees network connections per process, which is what dependency mapping requires. Agents are permitted, and it covers physical servers too.

D Migration Evaluator

Builds a business case and a TCO model. It answers what it will cost, not what connects to what.

THE SIGNAL

Two details decide it: undocumented dependencies, and a mix including physical servers, which agentless cannot reach. Module 33.

A board will not approve a migration programme without a comparison of current on-premises total cost against projected AWS cost, including Windows Server and SQL Server licensing options and more than one purchasing scenario. The company already holds three years of inventory and utilisation data from a third-party monitoring product.

The board meets in six weeks.

What should the solutions architect recommend?

- ☐ A The AWS Pricing Calculator, sized from the existing inventory
- ☐ B Application Discovery Service, then a cost model built in a spreadsheet
- ☐ C Migrate a pilot application and extrapolate its cost across the estate
- ☐ D Migration Evaluator, importing the existing third-party data, producing TCO with what-if purchasing scenarios

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Pricing Calculator
Pricing an AWS design. It does not know the on-premises cost, so it cannot produce the comparison the board asked for.
- B

 Discovery plus a spreadsheet
Rebuilds Migration Evaluator by hand, and a hand-built model carries little external credibility with a board.
- C

 Pilot and extrapolate
One application rarely generalises across a portfolio, and it will not complete within six weeks.
- D

 Migration Evaluator with imported data
Built for this artefact: TCO against current cost, multiple purchasing scenarios, and Windows and SQL licensing analysis. It accepts third-party data.

THE SIGNAL

"Comparison" is the word. Anything that prices only the AWS side answers half the question a board is asking. Module 33.

A team configured Migration Hub and selected us-east-1 as the home Region during setup. Workloads will migrate into eu-west-1 and eu-central-1, and company policy requires all migration metadata to be stored within the EU. Discovery data has already been collected into the existing home Region.

The programme starts in three weeks.

What should the solutions architect advise?

- ☐ A The home Region cannot be changed; use a different account and re-collect discovery data
- ☐ B Change the home Region to eu-west-1 in the Migration Hub console
- ☐ C Migration Hub data replicates automatically to the target Regions, so the policy is met
- ☐ D Configure a second home Region for the EU

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

 It cannot be changed
The home Region is set once per account and is permanent. Meeting the policy means a different account, and re-collecting discovery data.
- B

 Change it in the console
There is no such operation. This assumption is exactly what the question tests, and it is why the choice matters on day one.
- C

 It replicates
Migration Hub data stays in the home Region regardless of where migrations run. That is precisely the problem here.
- D

 A second home Region
One home Region per account, by definition. There is no second.

THE SIGNAL

An irreversible day-one choice made in a minute during setup, costing weeks to undo.
Module 33.

A data centre lease expires in nine months and will not be renewed. There are 920 servers running 240 applications. Every development team is committed to a regulatory programme for the next fourteen months and cannot take on migration work. Leadership has asked for a modernisation-first approach.

The applications currently work and are well understood.

What should the solutions architect recommend?

- ☐ A Refactor the applications to cloud-native services during the migration
- ☐ B Extend the data centre lease to create time for modernisation
- ☐ C Rehost the majority to meet the deadline, then modernise selected applications afterwards in AWS
- ☐ D Retain everything on-premises until development teams are available

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Refactor during migration

Refactoring needs developers, and there are none for fourteen months. Leadership's preference does not create engineering capacity.

C Rehost now, modernise later

Rehosting needs no application change, so it is unblocked by the developer shortage, and it is the only strategy that fits nine months at this scale.

B Extend the lease

The scenario says it will not be renewed. Changing a stated constraint is not an architectural recommendation.

D Retain everything

There will be no data centre in nine months. Retain requires somewhere to retain it.

THE SIGNAL

A preference meets a constraint, and the constraint wins. Saying so is the architect's job. Module 34.

Discovery across 1,600 servers finds 240 with no inbound or outbound network traffic over 90 days, no interactive logins, and no owner recorded anywhere. Several carry hostnames referencing projects that closed years ago. The migration budget assumes all 1,600 will move.

Some business processes run quarterly.

What should the solutions architect recommend?

- ☐ A Investigate the 240 for retirement: power off, retain recoverable for a defined window, then delete and remove from scope
- ☐ B Rehost all 1,600, since migrating is safer than deleting
- ☐ C Power off the 240 immediately and remove them from scope
- ☐ D Retain the 240 on-premises indefinitely

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Investigate, then staged retirement
Retire is the cheapest strategy. 15% leaves scope, and the recoverable window covers the quarterly processes 90 days of data would miss.
- B

 Migrate everything
Pays to plan, test, cut over and then run 240 servers nobody uses, indefinitely. Migrating waste preserves it.
- C

 Power off immediately
The right instinct without the discipline. The scenario says some processes run quarterly, which 90 days of observation can miss entirely.
- D

 Retain indefinitely
Keeps them powered on-premises forever. Retain is for workloads you will revisit, not ones nobody can identify.

THE SIGNAL

"Some business processes run quarterly" is planted to separate A from C. Ninety days is not a full business cycle. Module 34.

A company must vacate its data centre in five months. It runs 380 VMware virtual machines. Its infrastructure team is deeply skilled in vSphere and has no AWS experience, and retraining before the deadline is not realistic. Several applications are vendor-supplied and may not be modified.

The company intends to adopt native AWS services over the following three years.

What should the solutions architect recommend?

- ☐ A Refactor all 380 workloads to native AWS services now
- ☐ B Relocate to VMware Cloud on AWS to meet the deadline, then modernise into native services over time
- ☐ C Rehost all 380 with Application Migration Service and retrain the team in parallel
- ☐ D Retain the estate and negotiate a longer lease

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Refactor now
380 workloads refactored in five months, by a team with no AWS experience, including applications nobody may modify.
- C

 Rehost and retrain
The strong alternative. Rehosting meets the deadline, and it hands a vSphere-skilled team 380 EC2 instances to operate from day one.
- B

 Relocate, then modernise
The operating model is unchanged, so the vSphere team stays productive. Five months is achievable, and modernisation follows without a deadline.
- D

 Negotiate the lease
Changes the constraint rather than working within it, and the exit is presented as fixed.

THE SIGNAL

The team's vSphere skills separate relocate from rehost. Both meet the deadline; only one keeps the team productive. Module 34.

Following a commercial dispute, a company must move 420 servers from another public cloud provider into AWS within five months. The servers run a mix of Windows and Linux, and the applications may not be modified. The team believes AWS migration tooling only works from on-premises data centres.

Outbound internet access from the source environment is permitted.

What should the solutions architect recommend?

- ☐ A Export each server as an image, transfer it to S3, and import with VM Import/Export
- ☐ B Rebuild all 420 servers natively in AWS
- ☐ C Application Migration Service, installing the replication agent on the source servers in the other cloud
- ☐ D AWS DataSync to copy the servers into AWS

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Export and VM Import
A point-in-time copy, so the source keeps changing while you transfer, and cutover means a long outage per server.
- B

 Rebuild natively
The applications may not be modified, and 420 rebuilds in five months is not achievable regardless.
- C

 MGN from the other cloud
MGN works anywhere the agent reaches AWS on TCP 443 and 1500, including Azure, Google Cloud, Oracle Cloud and Alibaba. The team's belief is simply wrong.
- D

 DataSync
Moves file and object data. It does not migrate servers, operating systems or applications.

THE SIGNAL

The team's stated belief is the question. MGN is a cloud-to-cloud migration tool as much as an on-premises one. Module 35.

A weekend cutover of 55 servers failed. Eighteen applications would not start, reporting licence activation errors and unreachable database hosts. The team rolled back after eleven hours. MGN replication had reported healthy for six weeks, and no test instances had ever been launched.

The next cutover window is in four weeks.

What should the solutions architect recommend before the next attempt?

- ☐ A Extend the cutover window to a full weekend so there is time to resolve problems
- ☐ B Re-install the replication agents and restart replication from the beginning
- ☐ C Cut over the 37 applications that worked, and defer the 18 that failed
- ☐ D Launch test instances into an isolated subnet, resolve the licence and connectivity failures, and repeat until clean

◆ PAUSE AND DECIDE

What should the solutions architect recommend before the next attempt?

A Longer window

Turns an eleven-hour failure into a longer one. Debugging licence activation under cutover pressure is the worst possible moment.

C Defer the eighteen

Cuts over 37 servers that were also never tested. The same class of fault is waiting in them.

B Restart replication

Replication was healthy and is not the fault. Restarting discards six weeks of transferred data for no benefit.

D Test instances, iterate until clean

Non-disruptive and repeatable with the source still serving users. Hardware-bound licences and unreachable hosts are exactly what a test finds.

THE SIGNAL

"Replication healthy for six weeks" tells you the copying worked and the application never did. Those are separate questions. Module 35.

A team installed MGN replication agents on all 800 in-scope servers during the first month, to get replication established early. Migration waves run monthly, with the final wave completing in month eight. Finance queries an unexpected Application Migration Service charge that first appears in month four.

The team believed the service was free for the duration.

What is the explanation, and what should change?

- ☐ A The free period is 2,160 hours per server starting at agent installation; install agents per wave, shortly before each wave migrates
- ☐ B The charge is for staging EC2 and EBS, which cannot be avoided
- ☐ C The free period applies per account rather than per server
- ☐ D Request a service credit, since the service is advertised as free

◆ PAUSE AND DECIDE

What is the explanation, and what should change?

- A

 Free period starts at agent install
2,160 hours is 90 days from installation, so agents fitted in month one expire in month four. Install per wave and most servers stay free.
- B

 Staging infrastructure
Staging EC2 and EBS are charged, separately and from day one. A charge first appearing in month four is the free period expiring.
- C

 Per account
The free period is per source server, which is exactly what makes staggered installation worthwhile.
- D

 Request a credit
The terms were met. The service was free for 90 days per server, and the team consumed that on servers it was not yet migrating.

THE SIGNAL

Month four is the arithmetic: ninety days from a month-one install lands exactly there. That is the clue given. Module 35.

A regional facility must move 620 TB of archived imagery to Amazon S3. Its internet connection is 300 Mbps and is shared with operations. The facility closes in twelve weeks. The team has proposed copying the data over the internet using the AWS CLI with multipart upload.

The company is an existing Snow Family customer.

What should the solutions architect advise?

- ☐ A Proceed with the CLI, using multipart upload for parallelism
- ☐ B Do the arithmetic: at 300 Mbps the transfer far exceeds the deadline, so use Snow Family devices
- ☐ C Enable S3 Transfer Acceleration to speed up the upload
- ☐ D Use DataSync over the existing internet connection

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A CLI with multipart

Parallelism uses the link more efficiently. It does not make the link faster, and the link is the constraint.

C Transfer Acceleration

Improves throughput over long distances by reaching the AWS backbone sooner. It cannot exceed the facility's 300 Mbps.

B Arithmetic, then Snow

300 Mbps is about 3 TB a day at perfection, so 620 TB is roughly 200 days against an 84-day deadline. Three Snowball Edge devices cover it.

D DataSync

The best online tool available, and still bounded by the same 300 Mbps. Better software, same physics.

THE SIGNAL

Three options optimise the transfer; one asks whether it can finish. Note the scenario confirms existing Snow customer status. Module 36.

A media company must move 110 TB from an on-premises NFS share to Amazon FSx for Lustre, then keep the two synchronised for four months while applications cut over in waves. The site has a 10 Gbps Direct Connect link. Integrity must be verified for every file transferred. An engineer has proposed a nightly rsync job.

What should the solutions architect recommend?

- ☐ A The nightly rsync job as proposed
- ☐ B Snow Family devices for the initial 110 TB, then rsync for the ongoing sync
- ☐ C AWS DataSync, with scheduled incremental tasks and built-in integrity validation
- ☐ D S3 Transfer Acceleration into S3, then copy across to FSx

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

Nightly rsync
Rebuilds DataSync badly: no integrity validation, no managed retries, no throttling, and somebody owns the script for four months.
- B

Snow then rsync
Ships a box for data that transfers in under two days over the existing link, and still leaves the ongoing sync unsolved.
- C

DataSync, scheduled incremental
110 TB over 10 Gbps is a day or two. DataSync then handles four months of incremental sync, with validation and retries built in.
- D

Transfer Acceleration to S3
Adds an S3 hop the requirement does not need, and Transfer Acceleration does not target FSx.

THE SIGNAL

The same arithmetic that chose Snow in Q10 rules it out here. Compare the two: link speed decides, not data size. Module 36.

A company is moving a self-managed PostgreSQL 15 database from EC2 to Amazon RDS for PostgreSQL 15. The database contains 260 stored procedures, 55 triggers and several partitioned tables. A junior architect has scoped ten weeks for schema conversion using AWS SCT.

The database is 1.2 TB and downtime must be under 30 minutes.

What should the solutions architect advise?

- ☐ A Proceed with the ten-week SCT conversion as scoped
- ☐ B Rewrite the stored procedures for RDS compatibility
- ☐ C Use SCT in parallel with a full load plus CDC task
- ☐ D This is homogeneous, so no schema conversion is required; use a DMS homogeneous migration, which carries secondary objects

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

Ten-week conversion
Ten weeks converting PostgreSQL into PostgreSQL. The schema already runs on the target engine unchanged.
- B

Rewrite the procedures
They are already PostgreSQL procedures. RDS runs the same engine, so there is nothing to make compatible.
- C

SCT plus CDC
Correct on the data movement, and it still budgets conversion work this migration does not require.
- D

Homogeneous; no conversion
Same engine means the dialect is identical. DMS homogeneous migrations are serverless and carry procedures, triggers and partitions.

THE SIGNAL

Self-managed to managed is a hosting change, not an engine change. That single distinction saves ten weeks. Module 37.

A 26 TB SQL Server database must move to Amazon RDS for SQL Server. Restoring a native backup to the target takes about five hours, which the business accepts. However, the source will continue receiving writes for three weeks afterwards while an application is validated, and the two databases must stay in step throughout.

The engine and version are unchanged.

What should the solutions architect recommend?

- ☐ A A DMS full load plus CDC task instead of the native restore
- ☐ B Restore the native backup, then run a DMS CDC-only task for the three weeks
- ☐ C A DMS full load task, repeated nightly for three weeks
- ☐ D Restore the backup now, and restore it again at the end of the three weeks

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Full load plus CDC
Would work, and a DMS full load of 26 TB is slower than the native restore the business has already accepted.
- B

 Native restore, then CDC-only
CDC-only exists for exactly this: a target populated by other means, kept in step by DMS thereafter. Fastest path for 26 TB.
- C

 Nightly full loads
Reloads 26 TB every night for three weeks, and the target is stale for most of each day between runs.
- D

 Restore twice
Leaves the target three weeks out of date throughout validation, which is the requirement it fails.

THE SIGNAL

CDC-only needs a target already populated by something else. Combining native tooling with DMS replication is a real production pattern. Module 37.

A cutover performed at 01:00 updated a Route 53 record to point at AWS. By 10:00 the on-premises servers were still receiving about 55% of traffic, and the two environments had accepted conflicting writes. The record's TTL was 86,400 seconds.

The next cutover is in a month.

What should the solutions architect recommend?

- ☐ A Lower the TTL to 60 seconds immediately before the next cutover
- ☐ B Shut down the on-premises servers at cutover so traffic has nowhere else to go
- ☐ C Lower TTLs to 60 seconds at least a full TTL period before the next cutover, and keep the old environment answering during transition
- ☐ D Use a shorter maintenance window so divergence has less time to occur

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Lower it immediately before
The sharpest distractor. Resolvers caching the old record still hold 86,400 seconds, so the change is invisible for up to a day.
- B

 Shut down on-premises
Turns a divergence problem into an outage for 55% of users, and destroys the rollback at the same moment.
- C

 Lower TTLs a full period ahead
Lowering a TTL is itself subject to the old TTL, so it must happen at least one full period in advance to be live at cutover.
- D

 Shorter window
The window was not the problem. Traffic kept arriving for nine hours because of caching, not window length.

THE SIGNAL

Option A is the trap: lowering a TTL is itself a DNS change subject to the old TTL. That recursion makes it a planning task. Module 39.

A pricing service migrated to AWS in wave three now takes five seconds to return a quote, against 400 milliseconds before. It makes roughly 1,500 calls per quote to a rates database that remains on-premises. The Direct Connect link adds about 3 milliseconds each way. The team proposes upgrading the link from 1 Gbps to 10 Gbps.

What should the solutions architect recommend?

- ☐ A Migrate the rates database into the same wave as the pricing service, keeping the chatty pair together
- ☐ B Upgrade the Direct Connect link to 10 Gbps as proposed
- ☐ C Add a second Direct Connect connection for redundancy
- ☐ D Increase the pricing service's instance size

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Move the database with it
1,500 calls at 6 ms round trip is about 9 seconds of pure latency. Only removing the crossing fixes it, which means the pair moves together.
- B

 More bandwidth
Bandwidth is throughput, not latency. A wider link does not shorten the distance the packets travel.
- C

 Second connection
Resilience, not speed. Two links of the same latency give the same round trip.
- D

 Bigger instance
The service is waiting on the network, not short of CPU. A larger instance waits at the same speed.

THE SIGNAL

The arithmetic is handed to you: 1,500 calls times 6 ms. Bandwidth and latency are different properties. Module 40.

Where to revisit

IF YOU MISSED	REWATCH
Q1, Q2, Q3	Module 33 – Portfolio assessment and wave planning
Q4, Q5, Q6	Module 34 – The seven Rs
Q7, Q8, Q9	Module 35 – Server and application migration
Q10, Q11	Module 36 – Data migration
Q12, Q13	Module 37 – Database migration
Q14, Q15	Modules 39-40 – Cutover, and hybrid coexistence

ON THE EXAM

Compare Q10 and Q11: identical arithmetic, opposite answers. If you got one and missed the other, that is the lesson.